ECON 0150 | Economic Data Analysis

The economist's data analysis skillset.

Part 4.4 | The Problem of Timeseries

Part 4.3 Recap

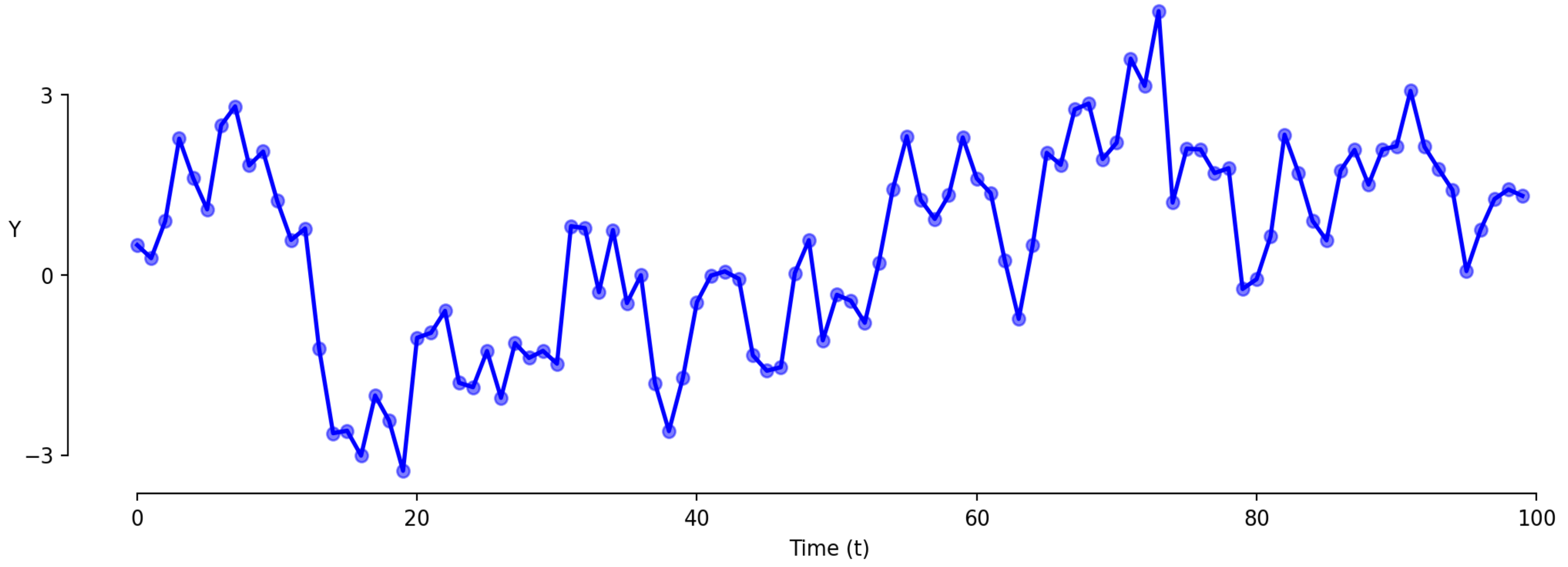
We check four assumptions to trust our model's results.

*1. **Linearity**: The relationship between X and Y is linear*
*2. **Homoskedasticity**: Equal error variance across all values of X*
*3. **Independence**: Observations are independent from each other*
*4. **Normality**: Errors are normally distributed*

*Today we focus on **Assumption 3: Independence**: the problem of timeseries.*

Timeseries Analysis

We often want to model relationships through time.



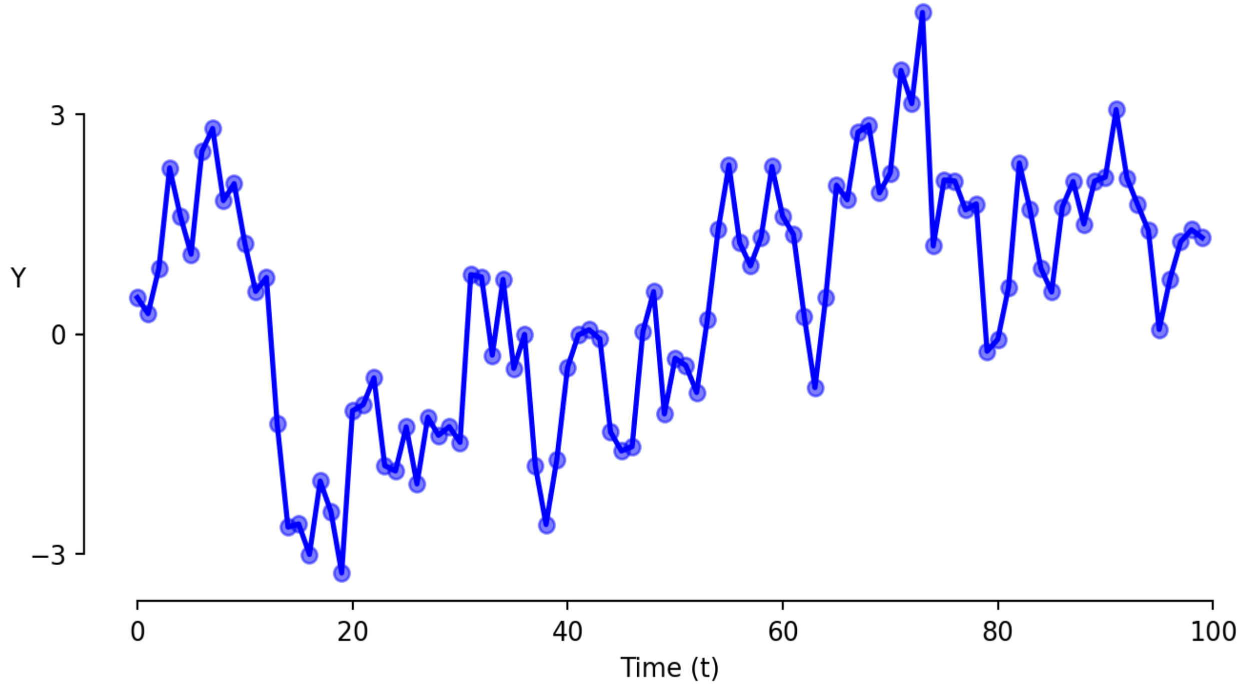
Data related in time has a special problem:

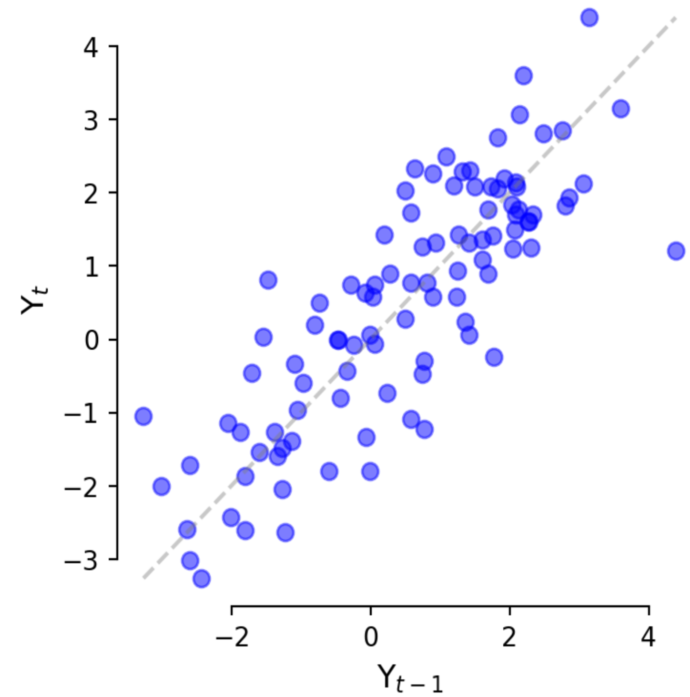
- *Observations are related to their past values (autocorrelation).*
- *This violates **Assumption 4: Independence**.*

Timeseries Analysis: Autocorrelation

We can check whether values in timeseries are related to their own past values.

A **Lag Plot** shows the value today (t) against the value yesterday ($t - 1$).



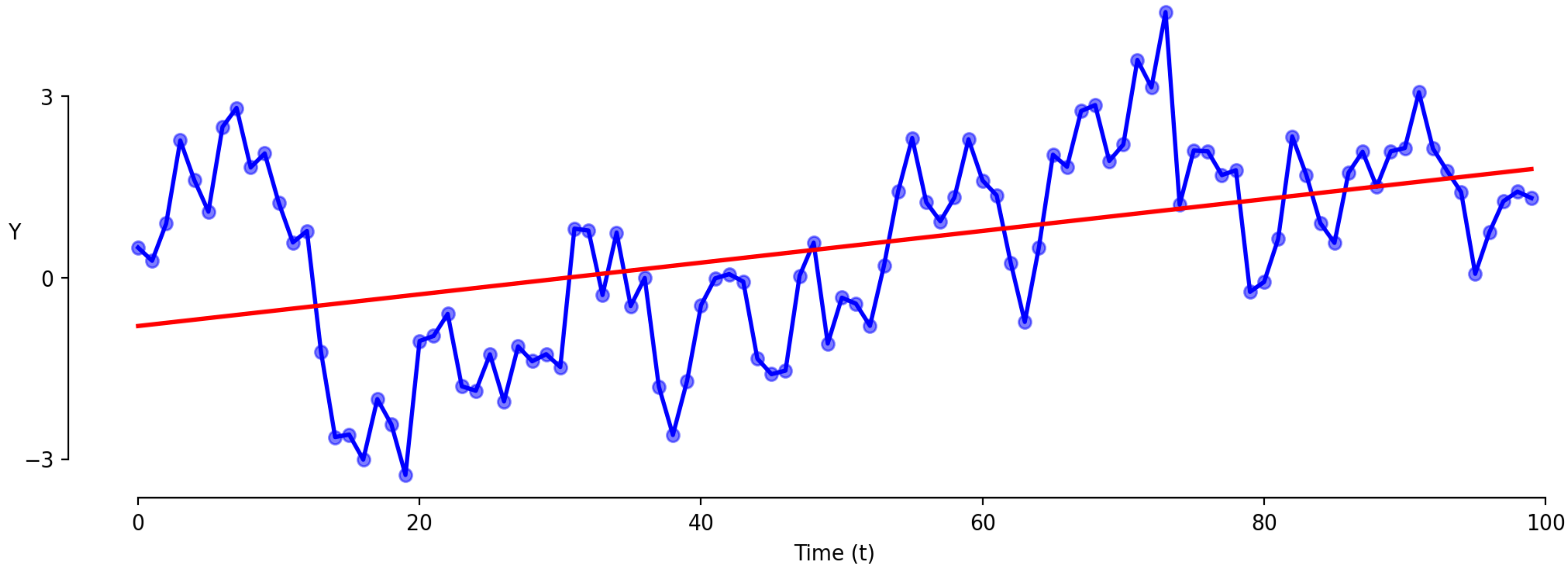


Autocorrelation tells us whether today's value depends on yesterday's value.

Timeseries: Model 1 (Levels)

The standard approach has problems with time series.

$$Y = \beta_0 + \beta_1 \cdot t + \varepsilon$$



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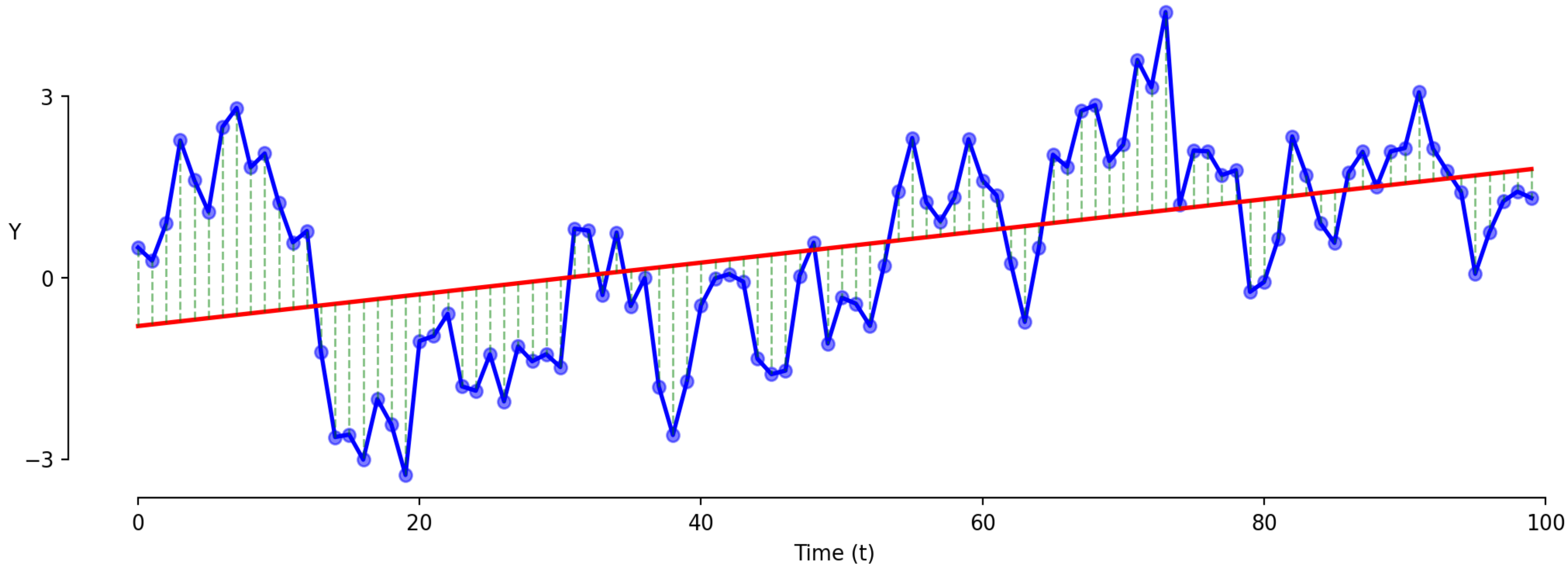
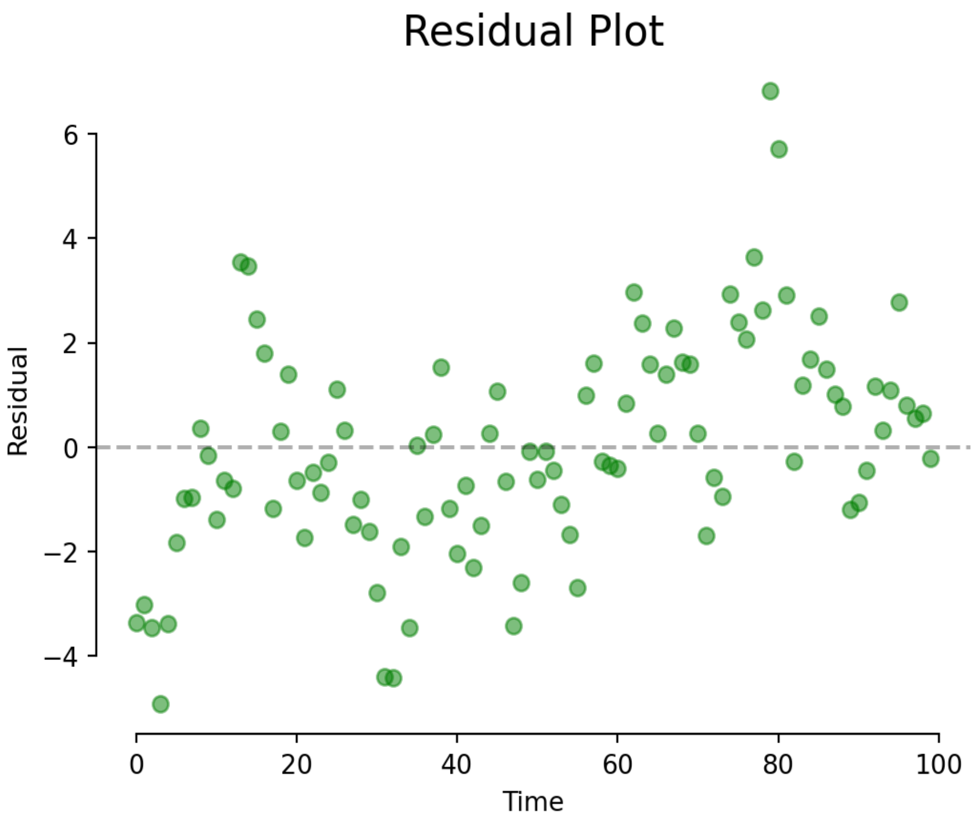


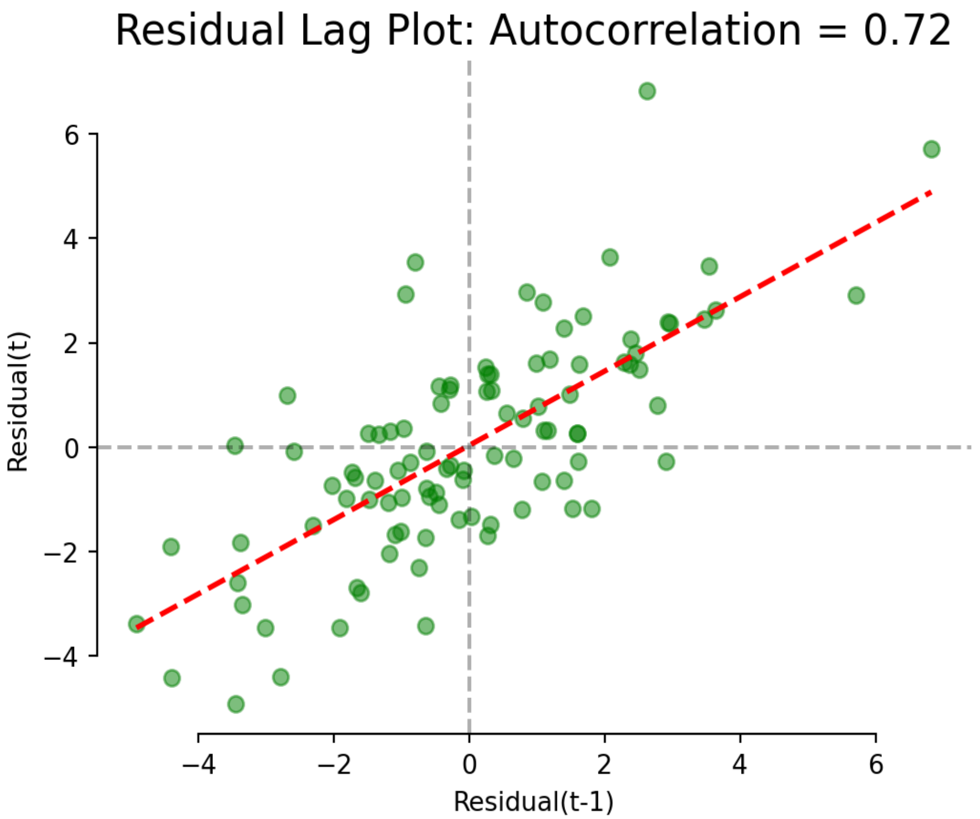
This model is often wrong in the same direction repeatedly.

Timeseries: Model 1 (Levels)

This 'levels' model shows strong patterns in residuals (autocorrelation).

$$Y = \beta_0 + \beta_1 \cdot t + \varepsilon$$





Exercise 4.4 | Model 1 (Levels)

Okun's Law: when unemployment rises, GDP tends to fall.

```python
# 1. Fit the levels model
model1 = smf.ols('gdp ~ unemployment', data=data).fit()
print(model1.summary().tables[1])
```

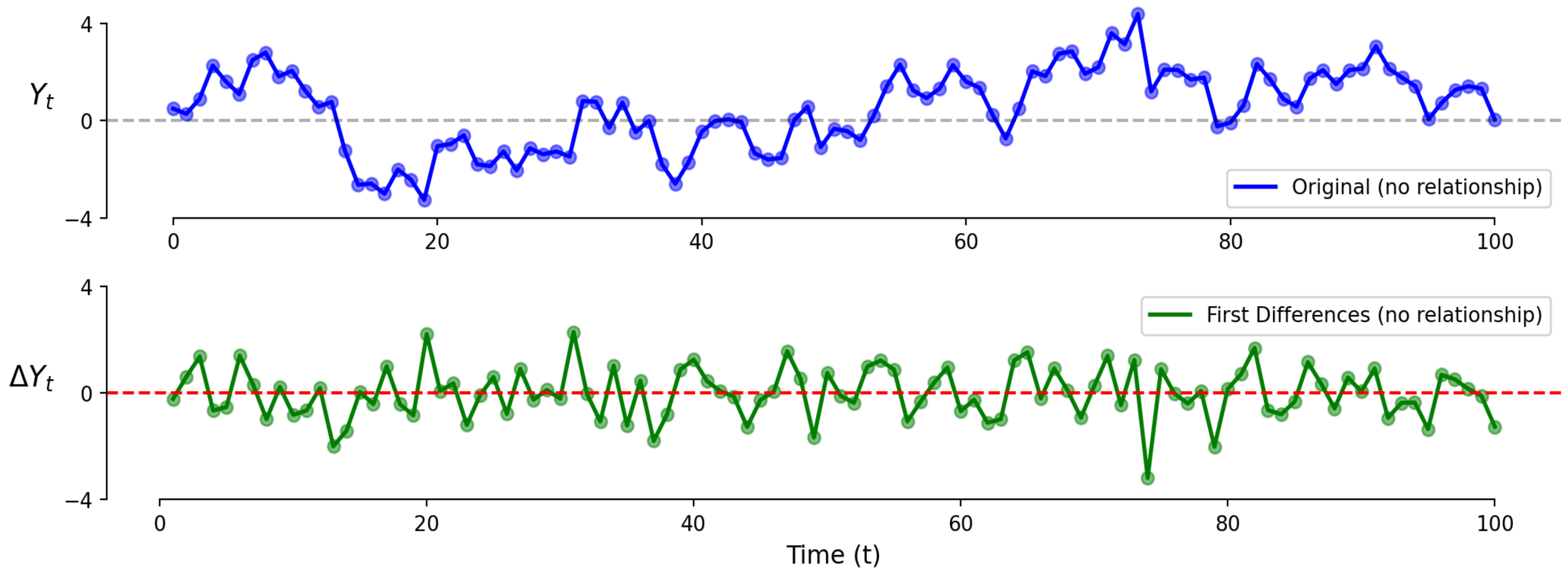
```python
# 2. Residual plot
plt.scatter(model1.predict(), model1.resid)
plt.axhline(0, color='red')
```

```python
# 3. Lagged residual plot
plt.scatter(model1.resid[:-1], model1.resid[1:])
```

Timeseries: Model 2 (First Differences)

We can fix some issues of autocorrelation by looking at changes instead of levels.

$$\Delta Y_t = Y_t - Y_{t-1}$$



Looking at changes instead of levels shows no relationship (correctly here).

What would first differences look like if there WAS a positive trend?

Timeseries: Model 2 (First Differences)

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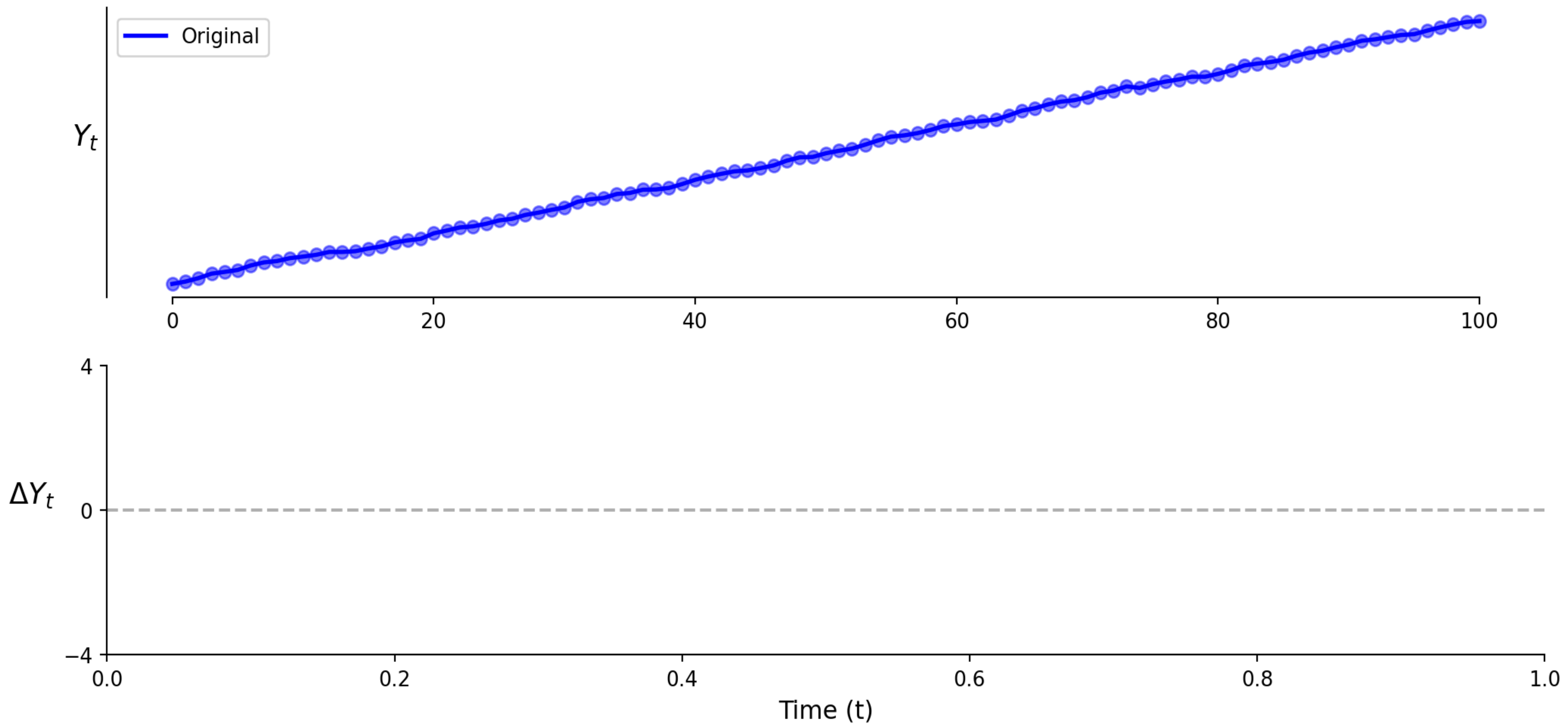

Timeseries: Model 2 (First Differences)

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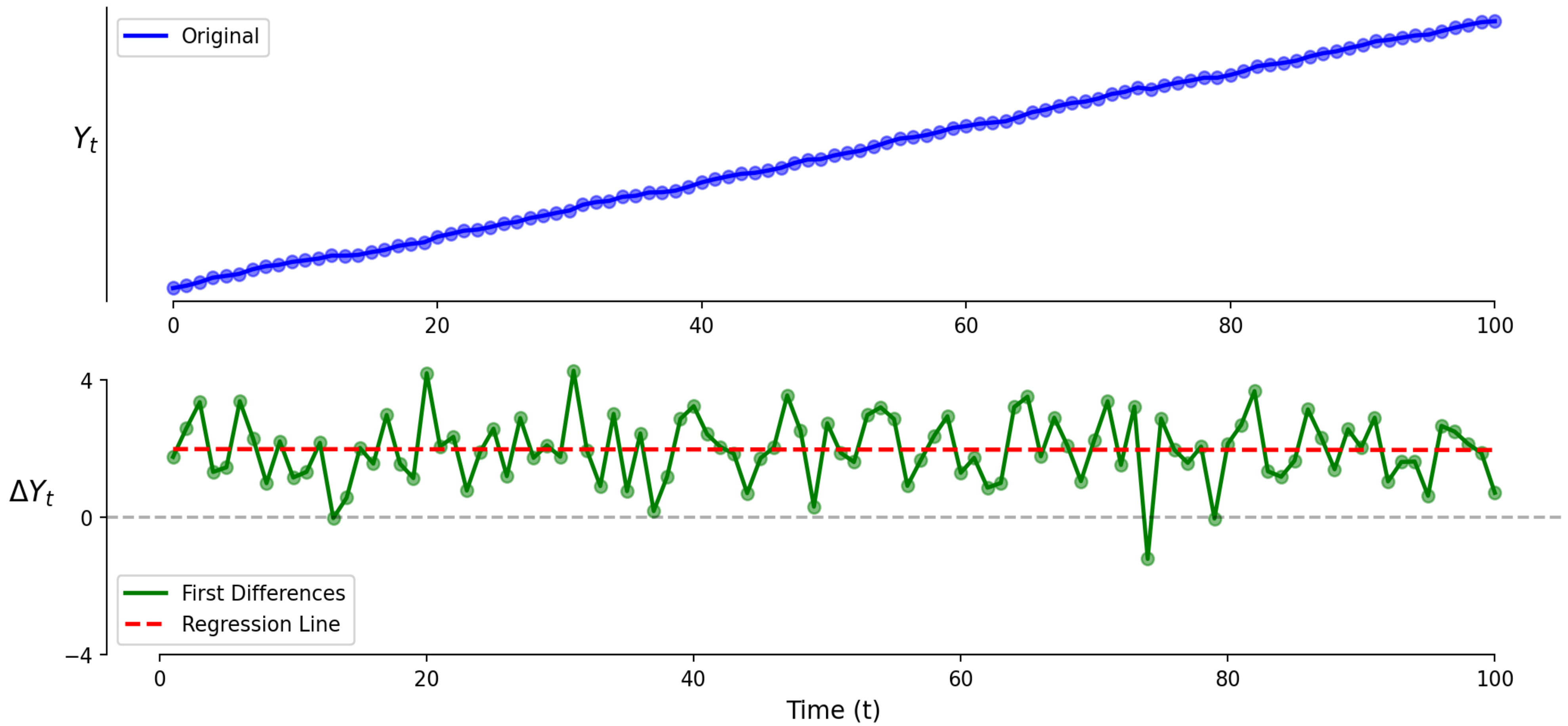
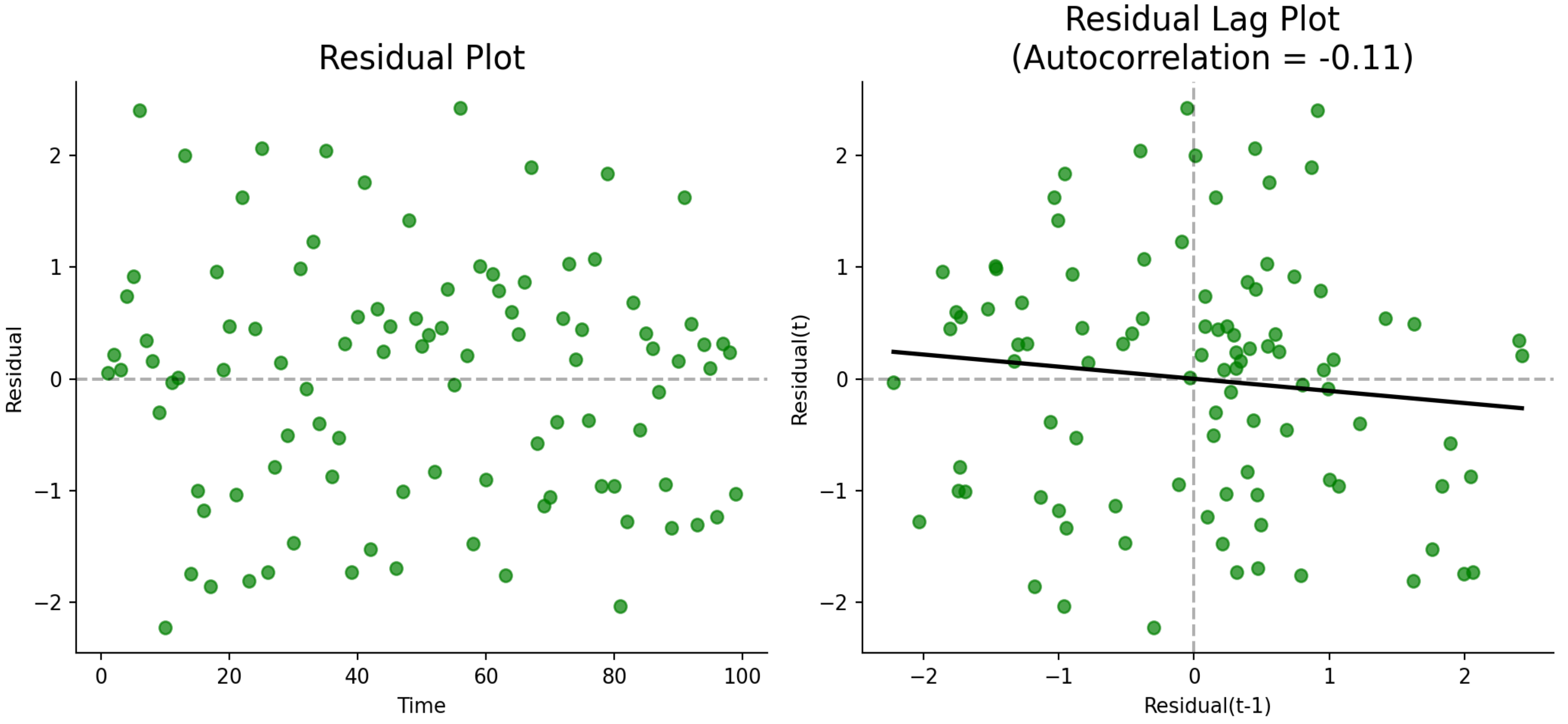


The vertical intercept β_0 is positive!

The slope coefficient β_1 is zero!

Timeseries: Model 2 (First Differences)

First differences reduces but does not eliminate the problem of autocorrelation.



Exercise 4.4 | Model 2

Examine a first difference model of the relationship between GDP and unemployment.

```
# Step 1. Create first differences of Y
data['gdp_diff'] = data['gdp'].diff()
data = data.dropna()
```

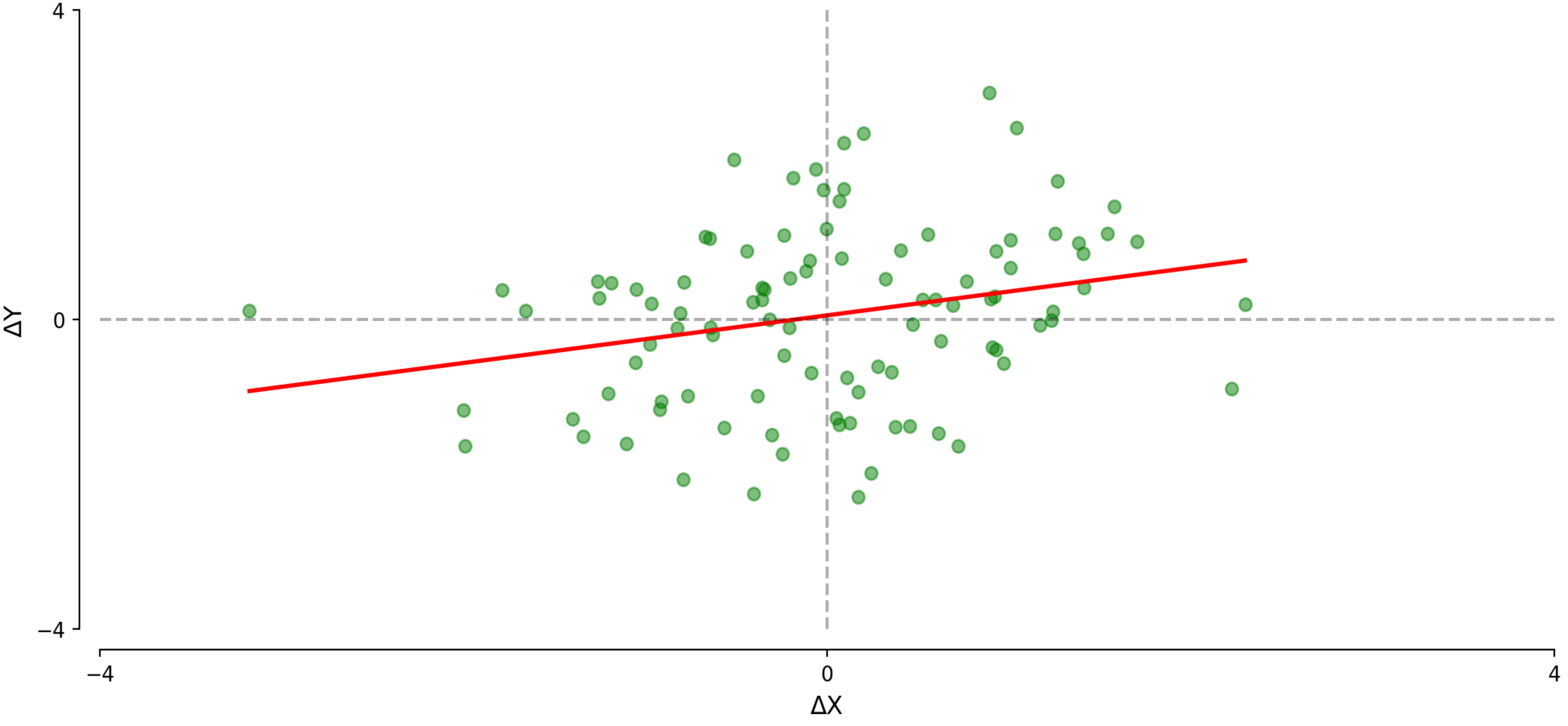
```
# Step 2. Fit the model (Y differenced, X in levels)
model2 = smf.ols('gdp_diff ~ unemployment', data=data).fit()
print(model2.summary().tables[1])
```

```
# Step 3. Residual and lagged residual plots
```

Timeseries: Model 3 (Double First Differences)

Sometimes we want to measure how two variables move together.

$$\Delta Y_t = \beta_0 + \beta_1 \times \Delta X_t + \varepsilon_t$$



Timeseries: Model 3 (Double First Differences)

Relating changes in X to changes in Y.

$$\Delta Y_t = \beta_0 + \beta_1 \times \Delta X_t + \varepsilon_t$$

- *Further reduces serial correlation in the error terms.*
- β_0 *captures time trend in Y*
- β_1 *captures the short-term relationship between variables.*
- *Clear interpretation: how do changes in X relate to changes in Y?*

Exercise 4.4 | Model 3

Examine a double first difference model of the relationship between GDP and unemployment.

```
# Step 1. Create first differences of X
data['unemployment_diff'] = data['unemployment'].diff()
data = data.dropna()
```

```
# Step 2. Fit the double differences model (both X and Y differenced)
model3 = smf.ols('gdp_diff ~ unemployment_diff', data=data).fit()
print(model3.summary().tables[1])
```

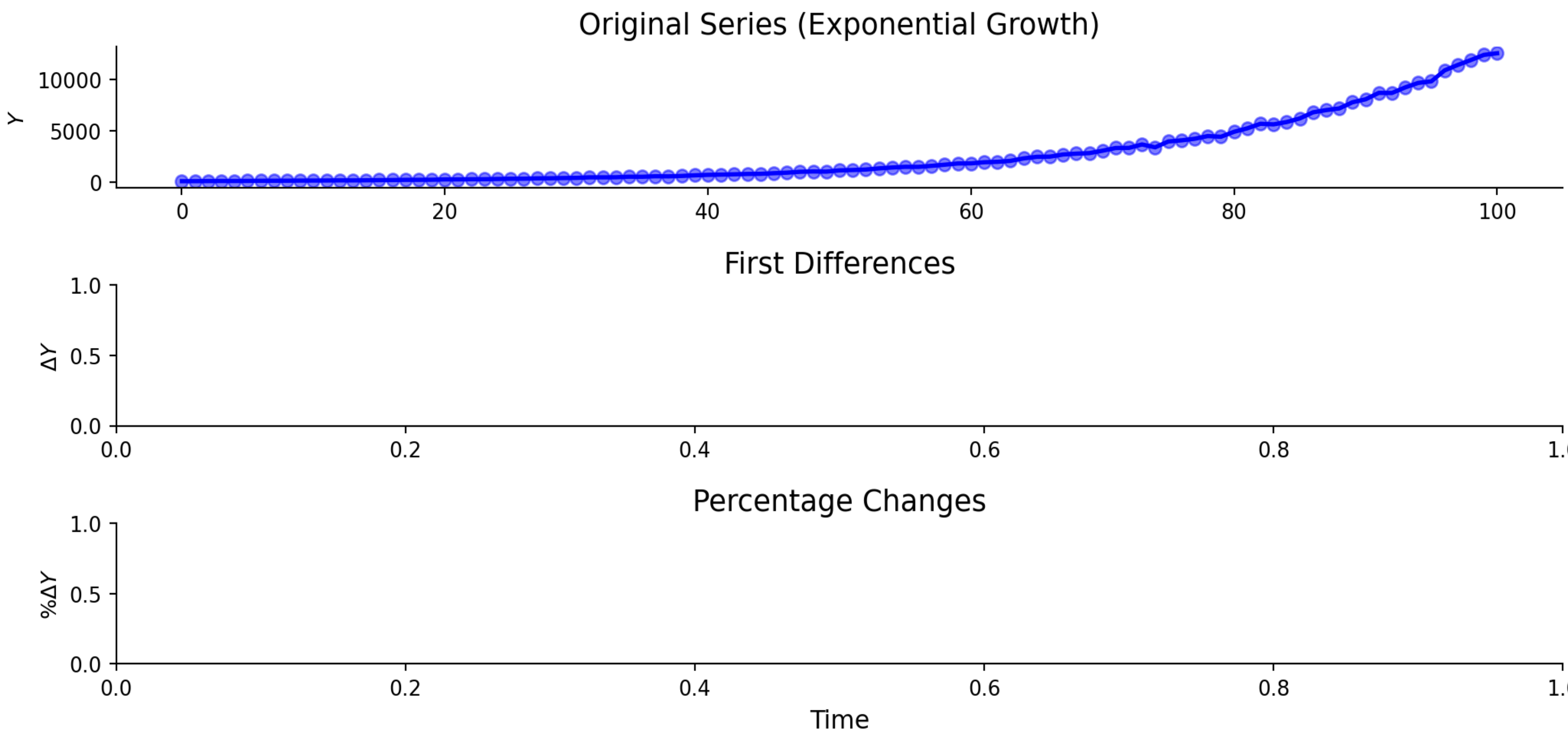
```
# Step 3. Residual and lagged residual plots
```

β_1 now represents the short-term relationship between changes in X and Y

Timeseries: Model 4 (Growth Rates)

Proportional changes provide interpretable coefficients:

$$g_Y = \frac{Y_t - Y_{t-1}}{Y_{t-1}} = \frac{\Delta Y_t}{Y_{t-1}}$$



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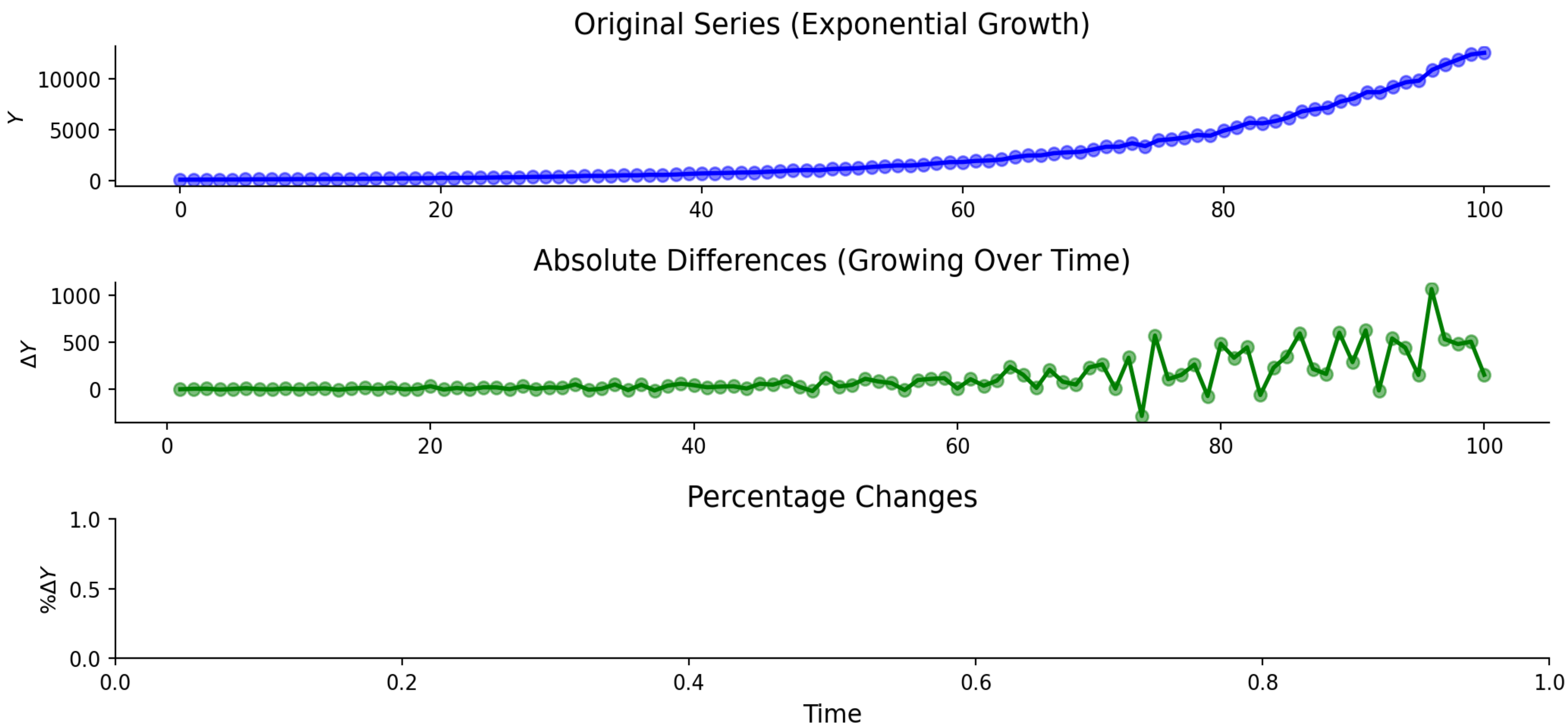

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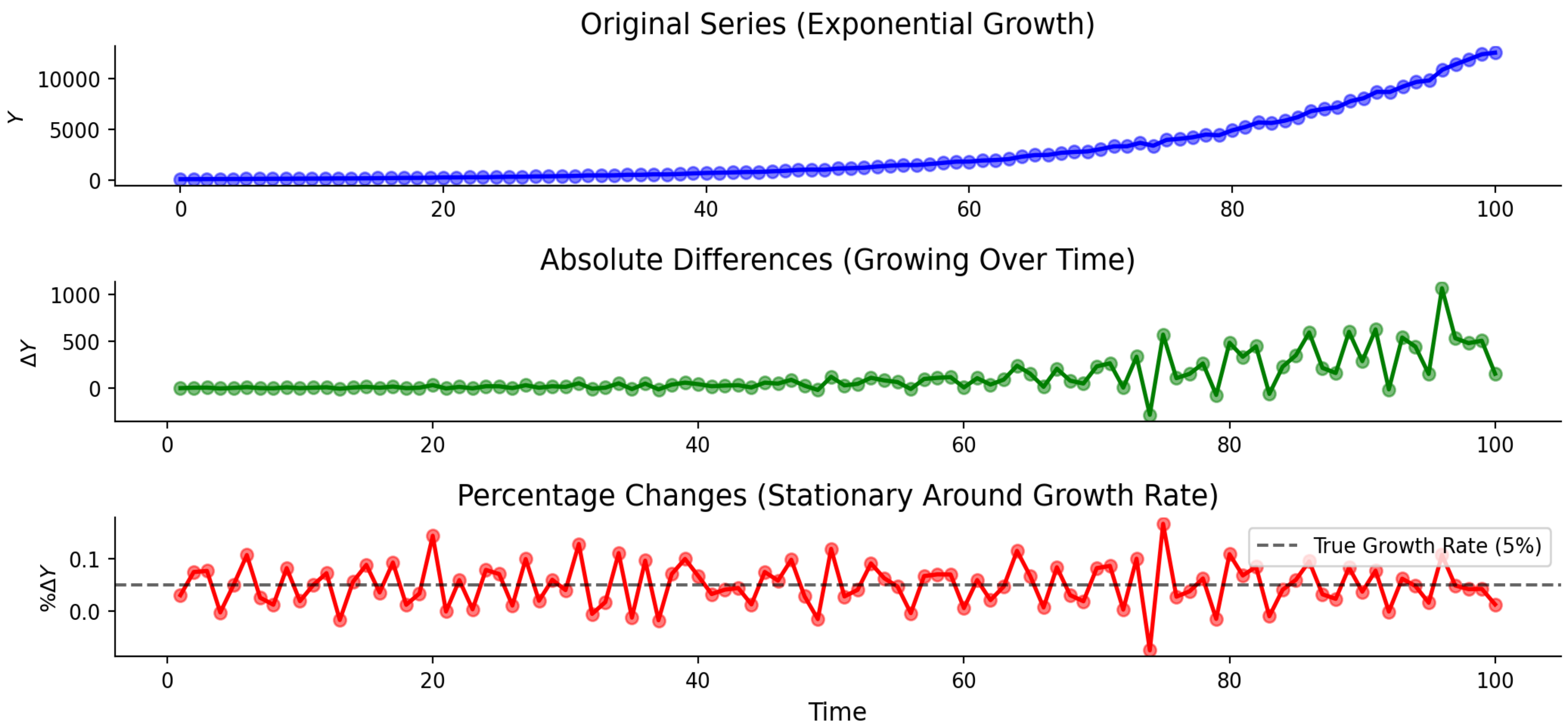
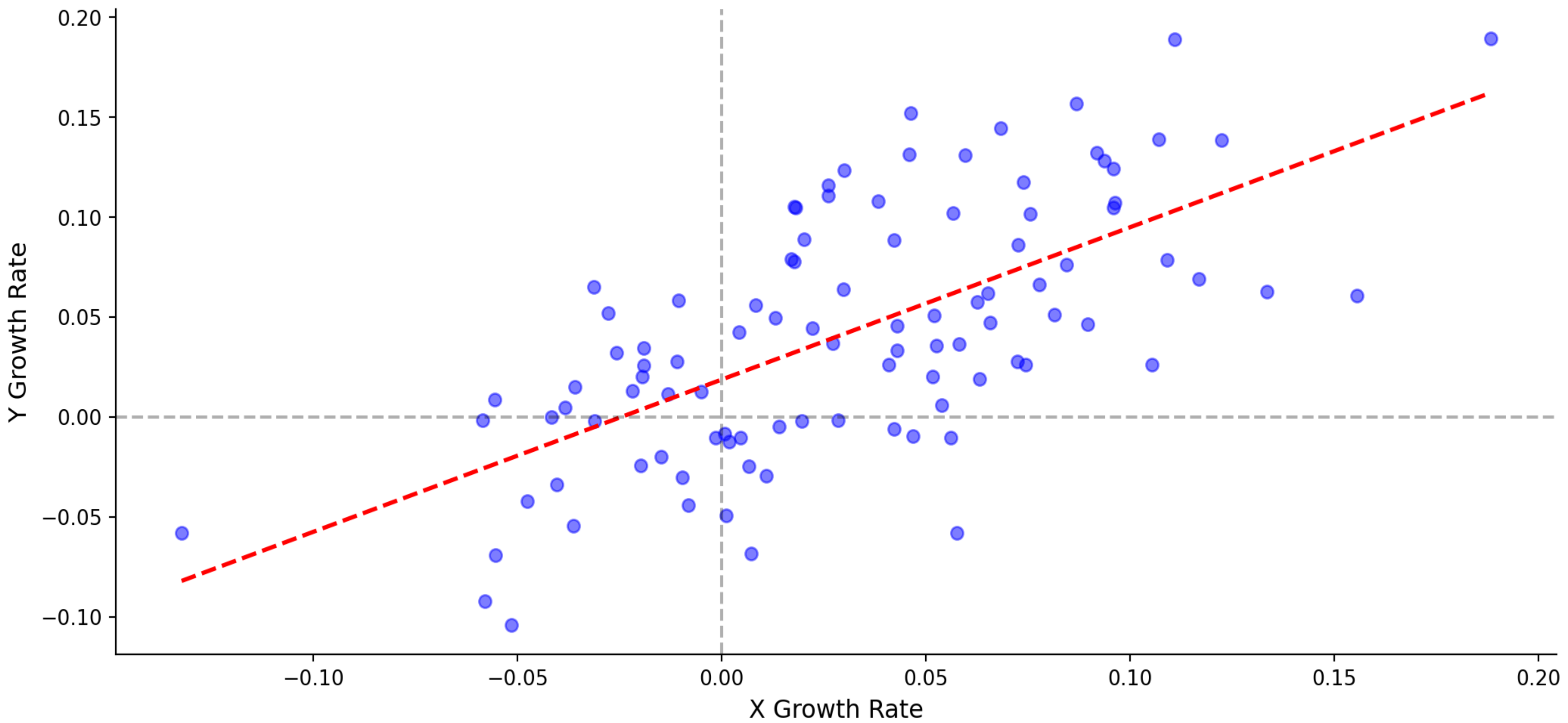


Timeseries: Model 4 (Growth Rates)

Is the growth in Y related to the growth in X?

$$g_Y = \beta_0 + \beta_1 \times g_X + \varepsilon_t$$



Timeseries: Model 4 (Growth Rates)

Is the growth in Y related to the growth in X?

$$g_Y = \beta_0 + \beta_1 \times g_X + \varepsilon_t$$

Growth rate models have the advantages of first differences and can scale better.

- *This is natural for variables with exponential growth.*
- β_0 *is Y's baseline growth rate.*
- β_1 *is how Y's growth responds to a 1 percentage point increase in X's growth.*

Exercise 4.4 | Model 4

Examine a growth rates model of the relationship between GDP and unemployment.

```
# Step 1. Calculate growth rates (percentage changes)
data['gdp_growth'] = data['gdp'].pct_change() # in percentage points
data['unemployment_growth'] = data['unemployment'].pct_change()
```

```
# Step 2. Drop rows with NaN values
data = data.dropna()
```

```
# Step 3. Fit the growth rate model
model4 = smf.ols('gdp_growth ~ unemployment_growth', data=data).fit()
print(model4.summary().tables[1])
```

```
# Step 4. Residual and lagged residual plots
```

β_1 *is now expressed in percentage point terms*

Timeseries: Summary

Autocorrelation can be reduced but not eliminated.

Model	**Equation**	**β_1 Interpretation**	**Best For**
Levels	$Y = \beta_0 + \beta_1 X + \varepsilon$	Relationship between levels	Cross-sectional data
First Diff	$\Delta Y = \beta_0 + \beta_1 \Delta X + \varepsilon$	Short-term relationship between changes	Trending time series
Growth Rates	$g_Y = \beta_0 + \beta_1 g_X + \varepsilon$	Response of Y's growth to X's growth	Exponentially growing series

Always check the residual lag plot. Differencing reduces autocorrelation but rarely eliminates it entirely.